

ATP group Badrory: A Critical Analysis of Techniques, Tactics, Procedures, and Comparative Research (COMP5003)

Submitted by:

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Introduction:

Kaspersky Worldwide Research Team recently identified the threat actor **Badrory** as the source of a massive APT campaign in 2023. The Advanced Persistent Threat (APT) group known as BadRory poses a serious cyber threat to Russian companies. This study provides a detailed examination of the group. This report aims to analyse BadRory's Tactics, Techniques and Procedures (TTPs) and suggest possible defence measures. BadRory's spear phishing attempts are aimed at a variety of organizations, including government agencies, defence companies and educational institutions. Badrory spied on and obtained all the information from government agencies in the Asia Pacific (APAC) area by sending spear phishing emails that contained malicious Microsoft Office documents that were encrypted by hardware. The phrase "Badrory" alludes to a passage from the well-known novel "The Adventures of Roderick Random," which follows the life of the title character Roderick "Rory" Random, who was taken out of one of the infected files. This literary allusion highlights the mystery and complexity surrounding this new threat actor while giving the group's identity more flavour.

Background:

Unlike other APT organizations, Badrory's country of origin is unknown, making it difficult to predict the APT's purpose and motivation. TTP can be compared to other ATP groups used, such as spear phishing. The APT BadRory group conducted two spear phishing operations against Russian companies. These attacks occurred between **October 2022 and April 2023**. Badrory attacked the target systems using spear-phishing emails that contained **Booby-trapped** Microsoft Office documents. In contrast to standard phishing, which has a wide and unfocused strategy, spear phishing targets particular people, companies, or positions within an organisation. His primary targets were Russian government agencies, military contractors, colleges, and hospitals. Speaking about some of the APT groups, most of them target military organizations to get and gather data on the targeted country from government bodies. Badrory may be a relatively new entrant into the world of cyber espionage, but his sophisticated activities indicate a high level of support and experience. The group's ability to successfully target respectable businesses suggests that it is a well-equipped and knowledgeable threat actor. But the organization's country of origin remains a mystery, raising questions about its goals and intentions. A group with an advanced level of skill and significant resources that allows him to create chances to achieve his goals using various attack methods such as physical, cyber and deception. Typically, these goals entail creating and strengthening footholds in the target organizations' IT infrastructure to steal information, disrupt or interfere with important components of the program, organization, or mission, and prepare themselves for the eventual achievement of these goals. An advanced persistent threat is one that consistently pursues its goals over an extended period, adapts to defence attempts to prevent it, and strives to maintain the level of interaction necessary to achieve its goals.



Fig 1: An illustration of the advanced persistent threat's (APT) life cycle, which repeats itself once it is finished.

Critical analysis:

Speaking about some of the APT groups, most of them target military organizations to get and gather data on the targeted country from government bodies. However, each appropriate organization will utilize a different target technique and malware, although spear phishing is the most common method. Spear phishing is a very specific type of phishing that aims to trick people or organizations into divulging private information. Various appropriate organizations employ various **TTPs (Tactics, Techniques, and Procedures)** Unlike other APT organizations, Badrory's country of origin is unknown, making it difficult to predict the APT's purpose and motivation. However, Badrory is a recently discovered APT group that has demonstrated its strength by attacking Russian government entities twice with spear phishing attacks. These attacks used booby-trapped Office emails between **October 2022 and April 2023**. The group's main targets were hospitals, universities, military contractors, and government agencies.

A recent Kaspersky investigation shows that he specifically targeted Russian universities, government organizations and the armed forces. This means that the main trolling technique used by BadRory is spear phishing via infected attachments or URLs. Unlike other APT organisations, BadRory has very simple TTPs. Badrory attacked the target systems using spear-phishing emails that contained booby-trapped Microsoft Office documents. This causes the system to become highly infected, which triggers the installation of Trojan. Sensitive data is usually stored on secure USB devices, commonly called encrypted USB drives, in a separate partition. All contents of this section are encrypted. Access to encrypted USB data requires a user password and software stored in the unencrypted portion of the USB device. Companies

and government organizations typically use these devices to store highly sensitive data. They then physically move the USB drive to a secure location and connect it to an isolated network.

According to Kaspersky, the APT group has created malware that will act like a worm and hide in the software of a secure USB drive, infecting every machine into which the device is inserted. Its main purpose is to remove files from the victim's computer and obtain access by issuing arbitrary instructions. APT BadRory's activities demonstrate a methodical and thoughtful approach to cyber warfare. The organization has effectively infiltrated sensitive targets, including government agencies, colleges, and military contractors, with very simple TTPs. Their success is due to their successful use of spear phishing, a tactic that exploits human weaknesses rather than systemic weaknesses. Notable is the group's emphasis on spear phishing. They trick users into opening booby-trapped Microsoft Office documents by creating convincing emails that include these documents, infecting their PCs. In addition to being cost-effective, this technique bypasses many conventional security procedures, which often place more emphasis on user behaviour than system weaknesses.

APT	TTP	Target sector
Mustang Panda	Using spear-phishing emails to spread the infection initially, Utilization of modified PlugX malware variants, using a malicious payload in a portable executable file (.exe), VBScript components in LNK files are used to automate collection and download more files	Non-government, government, and telecommunication
Space Pirate	Phishing emails are used to spread infection initially, Use of malware primarily comprised of Cobalt Strike, reverse shells, custom stagers, meterpreter, and the PlugX remote access trojan (RAT), Using the virus known as Shadow Pad	Government, aerospace, aviation, fuel and energy companies, educational institutions
Badrory	Using MS Office documents attached to spear-phishing emails to spread the initial infection Using a multi-level infection technique that installs a new Trojan, code injection into a USB drive's reputable access management program, which serves as a loader for malware on a new computer.	Government entities in the APAC region

Table 1: The above table displays the appropriate groups that targeted Russia's government and companies, and Badrory may have used some of these groups' identical TTPs.

Moreover, TTP BadRory's straightforwardness has its drawbacks. This facilitates a wide range of targets and ease of deployment, but with the necessary knowledge and training, it also makes it easier to identify and stop their strikes. By educating their members about the risks of phishing and the importance of verifying the source before accessing any documents or links, organizations can significantly reduce their risk. Another crucial component of Badrory's activities is the employment of malware that hides in the software of encrypted USB devices. They can breach isolated networks, which are normally more secure and difficult to access, thanks to this technology. Additionally, because the virus can re-infect computers each time a USB drive is inserted, it allows them to remain active on compromised systems.

Discussion:

Like other appropriate organisations, Badrory employed spear phishing emails with harmful content. Utilizing spear phishing emails with Microsoft Office documents, Badrory set himself apart from other similar organizations by encrypting the contents with hardware this demonstrates his level of technological expertise. Using this method, an organization can maximize data theft by not only hacking target systems but also remaining hidden for a long time. The long-term goal of the appropriate organisations is also highlighted in the discussion above. These groups' main goal is to spend as much time as possible inside the targeted system, steal as much data as they can, and intercept it. Furthermore, these appropriate groups are becoming a menace to international organisations and governments. The comparison of several APTs reveals variations in their place of origin, country of interest, industry, kind of malware utilized, and TTP. The goal of the apt groups is still the same despite all these variations: they want to break into the targeted organization's system and steal data in its cyber-attacks. In addition, BadRory stands out from other APT organizations due to its emphasis on hardware encryption, demonstrating deep knowledge of security systems and a desire to circumvent them. BadRory poses a serious threat to governments and international organizations due to this level of expertise. APT like Badrory in the globe indicates that we should have strong cybersecurity measures, be more vigilant against these types of attacks, and maintain a close watch on these sorts of assaults. Badrory's unknown origins make it hard to predict his motivations and goals, which exacerbates the tough circumstances and potential threats.

Conclusion:

In conclusion, BadRory, an Advanced Persistent Menace (APT) group, has become a serious cyber threat, especially to Russian companies. Although BadRory is a relatively new player in the world of cyber espionage, the group has proven to be very competent and successful in its activities. Spear phishing, its main attack method, targets specific individuals or organizations

by exploiting human weaknesses rather than system flaws. This method, along with the use of hardware-encrypted Microsoft Office documents, allows BadRory to penetrate target computers and remain hidden for an extended period. The organization's expertise in technology is further demonstrated by its ability to break into isolated networks using malware hidden in the software of encrypted USB devices. Unfortunately, there is a degree of mystery surrounding BadRory's origins, making it difficult to determine his goals and intentions. This incident emphasizes the need for strong cybersecurity defences and constant vigilance over these types of attacks. Protecting sensitive data and defending cybersecurity requires knowledge and the ability to combat entities like BadRory, as the cyber threat landscape is always changing.

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